CSE2023 ASSIGNMENT #6 (WILL NOT BE GRADED)

1) Find the solution to $a_n = 3a_{n-1} + a_{n-2} - 3a_{n-3}$ for n = 3, 4, 5, . . . , with $a_0 = 4$, $a_1 = 10$, and $a_2 = 12$.

2) Find the solution of the recurrence relation $a_n = 3a_{n-1} + 5 \cdot 3^n$.

3) A new employee at an exciting new software company starts with a salary of $2000 and is promised that at the end of each year her salary will be double her salary of the previous year, with an extra increment of $150 for each year she has been with the company.
   a) Construct a recurrence relation for her salary for her nth year of employment.
   b) Solve this recurrence relation to find her salary for her nth year of employment

4) Find the general solution for the following recurrence relations.
   a) $a_n = 2a_{n-1} + 2^n$

   b) $a_n = -6a_{n-1} - 9a_{n-2} + (-3^n)$

   c) $a_n = 3_{n-1} - 2a_{n-2} + 5$

   d) $a_n = a_{n-1} + 6a_{n-2} + 3^n$

   e) $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3} + 2^n$

   f) $a_n = 5a_{n-1} - 6a_{n-2} + 6(3^n)$